

Robust Pattern Generation for Small Delay Faults under the Impact of Variations

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Abstract—This work tackles the challenges of detecting Small Delay Faults (SDFs), which are exacerbated by timing variations caused by process, voltage, temperature, and aging. Many test patterns are required to ensure high fault efficiency under the impact of conditions introduced by these variation sources. However, having many test patterns to cover all conditions is practicality limited by excessive memory and execution time demands. This research focuses on proposing a testing methodology that enables test pattern generation under PVT for SDFs, minimizes memory overhead, and preserves and even improves fault efficiency.

I. INTRODUCTION

Many defects during manufacturing, operation or in the wear-out phase result in delay faults, particularly Small Delay Faults (SDFs). SDFs are characterized by an additional delay in a component that is smaller than the clock period. Despite their seemingly minor nature, SDFs are a major source of Silent Data Corruption, observed in large-scale systems like Google's Spanner and Meta's data centers, leading to data loss and increased system reliability risks [1], [2].

Current methods for detecting SDFs face significant challenges due to their dependence on the propagation of delays through the longest paths in a circuit. Timing-aware Automatic Test Pattern Generation (ATPG) attempts to exploit the timing information under nominal conditions to propagate defects via these longest paths [4]. However, timing-induced variations, mainly by process (P), voltage (V), temperature (T), and aging (A) cause the longest path to vary across different chip instances, which can invalidate test patterns generated for one specific instance when applied to another circuit instance. This variability makes it difficult to use a single test pattern set to detect SDFs across all instances, leading to the need for multiple test patterns to achieve comprehensive fault coverage. The authors of a recent work by SIEMENS highlighted these challenges in SDF detection as critical issues for future silicon life-cycle management [3].

To ensure robust fault detection, a test pattern set T' must account for the timing variations across all operating conditions $c \in \text{COND}$, where COND is determined by variations in process (P), voltage (V), temperature (T), and aging (A). The total number of possible conditions can be expressed mathematically as follows:

$$\text{COND} = P \times V \times T \times A$$

Even with simplified assumptions such as three process corners (slow, nominal, fast), three voltage levels (minimum,

nominal, maximum), and three temperature ranges (-40°C , 25°C , 125°C) and three aging stages (bring-up, operational, wear-out) result in 81 unique conditions, each requiring a dedicated set of test patterns. However, generating a test set for all these conditions is impractical even during the manufacturing phase due to the large test memory and computing time required by timing-aware ATPG [5].

This research aims to develop robust test methodologies under PVT variations by introducing a novel ATPG algorithm that addresses the limitations of conventional timing-aware ATPG for SDFs. The approach reduces memory requirements while maintaining—or even improving—fault coverage and efficiency, and it effectively shrinks the exploration space of conditional delay fault constraints (COND). The study promotes integrating robust testing under variations into manufacturing test, embedded test, and periodic in-field Deterministic Logic Built-In Self-Test (DLBIST).

II. ROBUST TEST SET GENERATION UNDER VARIATIONS

The proposed algorithm for robust test set generation adopts a statistical learning approach. This approach utilizes a Monte Carlo-based training set derived from various circuit instances to account for timing-induced variations. Similar to the k-Nearest-Neighbor (kNN) classification method, the training set serves as a model for generating SDFs patterns. The key assumption is that a fault-detecting pattern for one circuit instance will also detect the same fault in other instances with similar timing characteristics.

One of the primary challenges of this approach is the generation of test patterns for every circuit instance. As traditional timing-aware ATPG is computationally prohibitive due to the lack of parallelism in combinational test generation for SDFs, the approach uses statistical methods that combine timing-aware GPU-based fault simulation with non-timing-aware ATPG for transition faults. The proposed method is illustrated in Fig. 1.

A Monte Carlo-based training set C is produced by utilizing an accurate transistor model that accounts for variations. The initial test set T is generated by non-timing-aware ATPG for the transition fault model. The fault list F is generated by collecting the possibly detectable SDFs across all instances in C . Instead of relying on deterministic pattern generation, the robust test set is constructed iteratively. The main idea is to add new test patterns to T which are effective in detection of those undetected faults by current patterns in T across specified

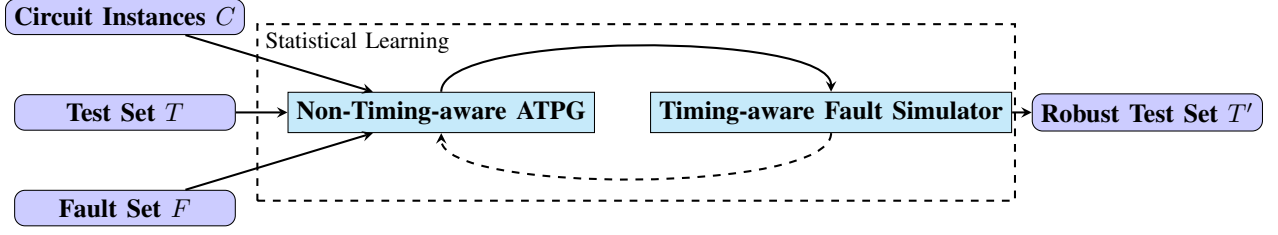


Fig. 1. Flowchart for Robust Test Set Generation using Statistical Learning

number of different predetermined SDFs sizes. The timing-aware fault simulator identifies faults not detected by the test pattern set T . These undetected faults are then sent back to the non-timing-aware ATPG for further new test pattern generation. This process continues until all faults are detected, or the predetermined fault sizes for SDFs are covered. As one fault may be detected at different sizes across different instances under variations, previous compaction methods become incapable of handling these varying fault sizes. A new timing-aware compaction strategy is developed to address this challenge by finding the minimum size at which each fault is detected, and then compacting the test even further.

This approach provides a new solution for robust test pattern generation under different variations, overcoming the limitations of conventional ATPG methods. The proposed algorithm generates test patterns for any $c \in \text{COND}$. The focus was then on reducing the COND space.

III. ACCELERATED PROCESS VARIATIONS AT LOW VOLTAGE

Testing across multiple voltages is essential, as certain defects, such as signal integrity issues and crosstalk, are only detectable at high voltage, while resistive bridges, gate-oxide defects, and resistive opens are primarily detected at low voltage. However, SDFs detection at low voltage is challenging due to amplified process variations, which mask fault effects. The proposed algorithm addresses this by generating patterns only for the voltage with the highest variations, with patterns for other voltages derived from this test set during compaction. This approach optimizes memory usage by increasing test set overlap across voltages.

IV. OPTIMAL VOLTAGE TESTING

Temperature fluctuations, particularly in automotive applications ranging from -40°C to 125°C , can cause significant timing variations, necessitating robust test sets. An optimal voltage was identified where temperature-induced variations are minimized. Test patterns at this voltage are smaller and improve fault efficiency due to reduced guardbands, thus accounting for temperature fluctuations while minimizing size and enhancing fault efficiency.

V. EXPECTED IMPACT

From a scientific perspective, the methodologies developed in this research will advance the state-of-the-art in delay fault

testing, particularly in the treatment of SDFs, which are a critical source of Silent Data Corruption. The proposed statistical learning-based approach for robust test pattern set generation represents a significant advancement by overcoming the limitations of traditional timing-aware ATPG methods, particularly in generating patterns for SDFs. By combining timing-aware GPU-based fault simulation with non-timing-aware ATPG, the proposed method offers the potential to generate test patterns under varying sources of variation, including process, voltage, temperature, and aging. The ability to handle complex, multi-dimensional variations will enhance fault detection efficiency, helping to mitigate Silent Data Corruption and ultimately result in more dependable systems, particularly in high-performance applications such as data centers and automotive systems. The proposed ATPG algorithm can be integrated into the manufacturing test flow. This work extends the robust test set by introducing a DLBIST scheme capable of executing tests under varying voltages. The proposed DLBIST structure aligns with current state-of-the-art architectures, requiring minimal modifications. Additionally, the method reduces the COND space by identifying conditions that subsume others, thereby minimizing complexity. This work proposes a multi-phase DLBIST scheme that enables robust testing during power-on, power-off, and operational phases under varying temperatures. Within this scheme, our method identifies an optimal voltage that minimizes temperature-induced variations, allowing generation of compact test patterns with high fault efficiency. By identifying subsuming conditions, the COND space is reduced, minimizing complexity. The approach suits DLBIST well, supporting periodic in-field testing at optimal conditions. From an industrial perspective, the approach reduces test pattern volume, memory requirements, and dependence on costly timing-aware ATPG. As a result, it lowers test costs, accelerates yield ramp-up, and enhances production efficiency for advanced integrated circuits.

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